

Document 17: IT Infrastructure & Wi-Fi Usage Policy

Institution: Hyderabad Institute of Technology (HIT)

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1. Campus Network Architecture and Authentication

The Hyderabad Institute of Technology maintains a state-of-the-art 10-Gigabit fiber optic backbone, delivering high-speed internet access across all academic blocks, the central library, and residential hostels. The university IT infrastructure is managed centrally by the Department of Information Services (DIS).

The campus wireless network is divided into two distinct broadcast networks (SSIDs):

- **HIT-SECURE:** This is the primary, encrypted enterprise network reserved exclusively for enrolled students and active faculty. Connection requires 802.1x authentication using the student's unique LDAP credentials (University Roll Number and a mandatory alphanumeric password).
- **HIT-GUEST:** This is an open, heavily restricted network designed solely for parents, visiting corporate recruiters, and guest lecturers. It requires OTP verification via an Indian mobile number. Students are strictly prohibited from using the guest network; the firewall will automatically block any student device attempting to bypass the secure network.

Students are permitted to register a maximum of three concurrent personal devices (e.g., one laptop, one smartphone, and one tablet) on the HIT-SECURE network. Attempting to authenticate a fourth device will force the system to automatically disconnect the oldest registered device.

2. Fair Usage Policy (FUP) and Data Quotas

While the university provides robust internet connectivity, bandwidth is a shared institutional resource. To prevent network congestion and prioritize academic research over entertainment, the DIS enforces a strict Fair Usage Policy (FUP) based on the student's academic tier.

- **Undergraduate Allocation (B.Tech):** Every undergraduate student is allocated a high-speed data quota of 50 Gigabytes (GB) per calendar month.
- **Postgraduate & Research Allocation (M.Tech/Ph.D.):** Recognizing heavier research requirements involving large datasets and cloud computing, graduate students are allocated 100 GB per calendar month.
- **Speed Throttling:** Base speeds on the HIT-SECURE network operate at up to 100 Mbps. However, the moment a student exhausts their monthly FUP quota, their bandwidth is instantly and automatically throttled to 2 Mbps for the remainder of the month. Data quotas reset at midnight on the first day of every new month.

Unused data does not roll over to the next month. Students who exhaust their quota cannot purchase "top-up" data packs from the university; they must manage their bandwidth responsibly or rely on their personal cellular data networks.

3. Prohibited Devices and Rogue Access Points

The university firewall actively scans the campus grid for unauthorized hardware that could compromise network security or cause IP address conflicts.

Students are strictly forbidden from bringing personal Wi-Fi routers, network switches, or LAN signal repeaters into the hostel rooms. Creating a "Rogue Access Point" by plugging a personal router into the hostel Ethernet port is a severe violation. The automated security grid will instantly detect the rogue signal, permanently block the wall port, and flag the room number for physical inspection.

Furthermore, the campus network is optimized exclusively for academic computing. The connection of smart home devices, IoT hardware not related to a verified academic project, Smart TVs, and dedicated gaming consoles (such as PlayStation 5, Xbox Series X, or Nintendo Switch) to the HIT-SECURE network is strictly blocked. If the firewall detects the MAC address signature of a gaming console attempting to authenticate, the student's account will be automatically suspended for 48 hours for violating hardware policies.

4. Cybersecurity Restrictions and Commercial VPN Ban

The university operates under strict legal compliance with the Ministry of Electronics and Information Technology (MeitY) guidelines. Consequently, all internet traffic passing through the HIT gateways is subjected to deep packet inspection.

Banned Activities:

1. **Peer-to-Peer (P2P) Torrenting:** The downloading or uploading of copyrighted movies, pirated software, or cracked games via BitTorrent protocols is actively blocked. Attempting to bypass this block triggers an automatic alert to the IT Helpdesk.
2. **Cryptocurrency Mining:** Utilizing university electricity and network bandwidth to run background crypto-mining scripts is classified as institutional resource theft.
3. **The Dark Web:** Accessing the Tor network or any .onion domains is entirely restricted, as these networks bypass institutional surveillance.

To enforce these rules, the university strictly bans the use of commercial Virtual Private Networks (VPNs) such as NordVPN, ExpressVPN, or Surfshark. Because VPNs encrypt traffic and hide it from the university firewall, their use is treated as an active attempt to circumvent campus security.

- **First Offense:** If a student is caught running a VPN on their laptop, their LDAP credentials are automatically disabled, resulting in a total network ban for 14 calendar days.
- **Second Offense:** A repeat violation results in a permanent ban from the HIT-SECURE network for the remainder of the semester, alongside a disciplinary fine of ₹5,000.

5. Computer Laboratory Conduct and Hardware Fines

The university maintains over a dozen specialized computer laboratories equipped with high-end workstations. These labs are strictly for scheduled academic practicals and approved project work.

- **Login/Logout Protocol:** Students must log into lab computers using their personal credentials. Leaving a workstation without officially logging off is a severe security risk. If a student leaves their account logged in and a subsequent user commits a cyber-violation from that terminal, the original student will be held 100% legally and financially liable.
- **Storage Restrictions:** Saving personal files, movies, or unapproved software to the local C-Drive of a lab computer is prohibited. All lab machines execute a deep freeze protocol at midnight, permanently deleting any files not saved to the designated secure network drive.
- **Hardware Vandalism:** The university has zero tolerance for hardware damage. Peeling off barcode asset tags, removing keycaps from mechanical keyboards, or deliberately severing mouse cables is classified as vandalism. If caught via the lab CCTV cameras, the student will be fined a flat rate of ₹2,000 for peripheral damage, or ₹15,000 if the monitor or CPU chassis is damaged.

6. Digital Identity and University Email Usage

Upon admission, every student is issued an official institutional email address (formatted as *rollnumber@student.hit.edu.in*) hosted on the Google Workspace for Education platform. This email is the sole channel for all official administrative communication, examination alerts, and placement drive links.

The institutional email is an academic tool, not a personal asset. Students are strictly prohibited from using their @student.hit.edu.in address to register for commercial streaming services (like Netflix or Spotify), online dating platforms, or politically affiliated forums. If the university's automated spam filters detect heavy influxes of promotional emails from non-academic commercial entities, the student's inbox will be temporarily frozen.

Six months after the student successfully graduates and receives their degree, the IT department will initiate a permanent purge of their institutional email account and associated Google Drive storage. Students are responsible for migrating their resumes, project files, and contacts to a personal Gmail account before this deadline. The only exception is for founders operating under the "Alumni Entrepreneur Extension" at the HIT Incubation Center, who are granted a one-year extension on their email privileges.